

Calculations for Greywater treatment

Calculations for Previous Hybrid Algal Bioreactor System Design

IMPORTANT NOTE: These calculations pertain specifically to the previous conceptual design of the Hybrid Algal Bioreactor System, which featured separate Flat Panel Photobioreactors and a distinct Airlift Column.

For the refined design, new calculations for bioreactor volume distribution, component dimensions, and pump sizing are required, as the physical configuration and integration are fundamentally different. These calculations are provided here for context and to show the evolution of the design process.

Goal: To change key calculations from initial overall wastewater treatment to now specifically focused greywater treatment.

Original Assumption: 2,400 Liters/day (for 16 people, assuming 150 L/person/day total wastewater).

Research: Percentage of greywater is about 65% to 70% and about 36 Liters per capacity per day.

Calculation: 36 Lpcd x 4 people = 144 L/day (1 house, 4 people, 36 Lpcd greywater generation).

Refinement of Influent Greywater Characteristics (N, P, NH_4^+)

Objective: To establish realistic and context-specific influent concentrations for Nitrogen (N), Phosphorus (P), and Ammonium ($\text{NH}_4^+\text{-N}$) in light greywater for a conceptual household-level bioreactor serving low-income communities in rural/peri-urban Maharashtra, India. This refinement moves beyond general wastewater assumptions to a more targeted design.

Previous Assumptions (from initial broad scope):

- Influent Phosphorus (P_{in}): 7 mg/L (Total P)
- Influent Nitrogen (N_{in}): 40 mg/L (Total N) (These were based on general domestic wastewater characteristics, not specific to greywater or the target region.)

Refining Greywater Characteristics: A Context-Specific Approach to Nutrient Inputs

My initial project scope used generalized wastewater parameters as a preliminary basis. However, to enhance the realism and applicability of this conceptual bioreactor design for its intended context - low-income, peri-urban, and rural communities in India - a more precise characterization of greywater influent was deemed essential. This refinement underscores the

iterative nature of engineering design, where initial assumptions are continually validated and adjusted with more specific data.

To achieve this, I conducted targeted literature searches focusing on "greywater composition low-income India," "nutrient concentration greywater Maharashtra India," and "ammonium concentration greywater India." Key findings from this review were analysed:

- **Source 1.2 (Sinnar, Nashik, Maharashtra):** This source provided specific data for a rural residential area in Maharashtra, detailing Nitrite (0.08 mg/L), Nitrate (0.67 mg/L), and Phosphate (0.012 mg/L) concentrations. While valuable for its regional specificity, the reported phosphate value appeared notably low for typical total phosphorus content in greywater.
- **Source 3.4 (General Greywater Characterization Table):** This source presented broader average untreated greywater characteristics, including Total Nitrogen (N) at 18.98 mg/L and Total Phosphorus (P) at 0.330 mg/L. These values offered a more comprehensive representation of typical greywater composition.
- **Source 3.2:** This publication, while not providing concentrations, indicated significant daily loads of N (45g) and P (3g) from average greywater volumes, reinforcing the substantial presence of these nutrients.

Decision & Justification for Revised Influent Concentrations

Based on this comparative analysis, I have updated the influent concentrations for my conceptual design to more accurately reflect greywater characteristics within the target context:

1. Influent Phosphorus (Pin): 0.330 mg/L (Total P)

- a. **Reasoning:** The phosphate (PO_4) concentration from the Maharashtra-specific study (0.012 mg/L) was considered unusually low for total phosphorus. To ensure the conceptual design's robustness and capacity to manage a more representative greywater load, the **0.330 mg/L Total Phosphorus (TP)** value from Source 3.4 was adopted. This value provides a more generalized average for untreated greywater in India and serves as a conservative (higher) design basis compared to the localized data, thereby ensuring sufficient bioreactor capacity.

2. Influent Nitrogen (N-in): 18.98 mg/L (Total N)

- a. **Reasoning:** Similarly, the nitrite and nitrate values from the Maharashtra-specific source (Source 1.2) were relatively low and represented only specific forms of nitrogen. Therefore, the **18.98 mg/L Total Nitrogen (TN)** value from Source 3.4 was selected as a more comprehensive and representative average for total nitrogen in untreated greywater in India. This ensures the design accounts for the full spectrum of nitrogen forms requiring removal or transformation by the algal system.

3. Ammonium ($\text{NH}_4^+\text{-N}$) Concentration:

- a. **Reasoning:** Direct average data for Ammonium-Nitrogen ($\text{NH}_4^+\text{-N}$) in greywater for low-income Indian communities was not consistently available

across the reviewed sources. However, as ammonium is a preferred nitrogen source for algal growth (requiring less energy for assimilation), and greywater contains organic matter (e.g., from soaps, skin cells) prone to biodegradation into ammonium, it is a reasonable **working assumption for this conceptual design** that a significant portion of the Total Nitrogen (18.98 mg/L) will be in forms readily available or convertible to ammonium for *Chlorella vulgaris* uptake. This assumption directly supports the anticipated biological efficiency of the algae.

Impact of These Changes:

These refined influent concentrations, combined with the updated daily greywater flow (144 L/day for a single household of 4 people), will now be utilized to recalculate the bioreactor volume, expected biomass production, and CO₂ capture potential. This iterative refinement process is critical in engineering design, leading to a more accurate, context-specific, and ultimately, more viable conceptual solution.

Hydraulic Retention Time (HRT): My Approach to Ensuring Effective Treatment

For this conceptual bioreactor, I've settled on a Hydraulic Retention Time (HRT) of **6.5 days**. My thinking here was to provide ample time for the algae to do their work. I initially chose this based on research suggesting that a 6-7 day HRT is often ideal for achieving really efficient Total Nitrogen (TN) removal in similar algal wastewater treatment systems. It's a sweet spot that allows enough contact time for the biological processes to unfold effectively.

Now, it's fascinating to consider how different nutrients behave within this 6.5-day window:

- **Phosphorus (P) Removal:** What's interesting is that algae are incredibly efficient at taking up phosphorus, especially in its orthophosphate form. My calculations consistently show that the volume needed for phosphorus uptake alone (around 1.0 Liter) is tiny compared to the overall bioreactor volume (936 Liters) set by the HRT. This tells me that phosphorus removal is a much faster process, and it's definitely **not the limiting factor** for how big my bioreactor needs to be or how long the water stays in it. The 6.5-day HRT gives the algae plenty of time for near-complete P removal.
- **Total Nitrogen (N) Removal:** Nitrogen is a bit more complex, biologically speaking. It involves the algae directly taking up forms like ammonium and nitrate but also relies on other microbial processes that convert organic nitrogen into these usable forms. That's why a longer HRT, like my 6.5 days, is so crucial here. It ensures there's enough time for all these transformations and for the *Chlorella vulgaris* to assimilate the nitrogen effectively, helping me hit that 95% removal target.
- **Ammonium (NH₄⁺-N) Preference:** Algae generally prefer ammonium as their nitrogen source because it's energetically more favorable for them to use. This is a real advantage for my chosen HRT. If a good portion of the Total Nitrogen in the

greywater is ammonium, the 6.5 days should be more than sufficient for efficient uptake. Even if there's a lot of organic nitrogen, this HRT provides the necessary time for those organic compounds to be broken down into ammonium by the bacteria that naturally coexist with the algae.

In essence, my 6.5-day HRT is a holistic choice for the entire greywater treatment process. It's designed to be robust enough to handle the slower, more complex biological steps (like comprehensive Total Nitrogen removal and building up enough algal biomass), while easily accommodating the faster processes like phosphorus and direct ammonium uptake. This way, I'm optimizing the design to achieve high removal efficiency across all the key nutrients.